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PLANAR INDUCTOR STRUCTURE WITH INTEGRATED SHIELDING TO CONTAIN ELECTRIC FIELD

Abstract

An example planar inductor includes a winding structure including a plurality of conductive windings arranged in layers of a printed circuit board (PCB) around a core, where the plurality of conductive windings include at least a top winding and a bottom winding. The planar inductor includes a conductive shield encasing the winding structure. The conductive shield includes an upper conductive shield layer disposed over the winding structure in a first direction, a lower conductive shield layer disposed under the winding structure in a second direction opposite the first direction, an inner conductive shield disposed within and around the plurality of conductive windings, and an outer conductive shield disposed around and outside the plurality of conductive windings.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of and priority to U.S. Provisional Patent Application No. 63/556,982, filed Feb. 23, 2024, entitled “PLANAR INDUCTOR STRUCTURE WITH INTEGRATED SHIELDING TO CONTAIN ELECTRIC FIELD,” the entire content of which is hereby incorporated herein by reference in its entirety.

BACKGROUND

[0002] By adopting wide-bandgap (WBG) power devices using silicon carbide (SiC) or gallium nitride (GaN) material, switch-mode power converters are able to achieve high power density while maintaining good efficiency. Compared to Si power devices, these WBG devices experience reduced switching losses, enabling a higher switching frequency and the use of smaller passive energy storage components within the converter. However, the use of high switching frequency, along with the fast turn-on and turn-off speed of WBG devices, can worsen the electromagnetic interference (EMI) emissions generated by power converters. To ensure compliance with electromagnetic compatibility (EMC) regulations, suppressing the EMI emissions generated by power converters while achieving the desired power density and efficiency is important.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Many aspects of the present disclosure can be better understood with reference to the following drawings. The components in the drawings are not necessarily to scale, with emphasis instead being placed upon clearly illustrating the principles of the disclosure. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views.

[0004] FIG. 1 depicts a cross-sectional schematic of an example planar inductor implementing conductive shielding using shield layers and conductive edge plating according to various embodiments of the present disclosure.

[0005] FIG. 2 depicts a cross-sectional printed circuit board (PCB) schematic of an example inductor structure implementing the conductive shield shown in FIG. 1, according to various embodiments of the present disclosure.

[0006] FIG. 3 depicts a perspective view of a planar inductor implementing the inductor structure shown in FIG. 2 according to various embodiments of the present disclosure.

[0007] FIG. 4 depicts a cross-sectional schematic of an example planar inductor with an alternative implementation of a conductive shield using conductive shield rings located co-planar with inductor windings, according to various embodiments of the present disclosure.

[0008] FIG. 5 depicts a cross-sectional PCB schematic of an example inductor structure implementing the conductive shield shown in FIG. 4 according to various embodiments of the present disclosure.

[0009] FIG. 6 depicts a perspective view of a planar inductor implementing the inductor structure shown in FIG. 5 with a view of an alternative implementation of the conductive shield layers using patterned conductive features according to various embodiments of the present disclosure.

[0010] FIG. 7 depicts a sectional view of the planar inductor shown in FIG. 6 including inductor windings, conductive shield rings, and conductive patterned shield layers according to various embodiments of the present disclosure.

[0011] FIG. **8** depicts a planar inductor with a patterned conductive shield implemented with a UI shaped core according to various embodiments of the present disclosure.

DETAILED DESCRIPTION

[0012] Power converters in many applications utilize buck, boost, or buck-boost topologies, and other topologies are known. In these converters, the power inductor is often a primary component that adversely affects electromagnetic compatibility (EMC). This is attributable to the strong electric and magnetic fields generated by power inductors. These fields couple into other areas of the circuit, forming parasitic inductive and capacitive couplings which generate conducted EMI noise. In particular, common mode (CM) conducted noise is increased when the electric field from the inductor forms capacitive couplings with other areas of the converter circuit.

[0013] According to the embodiments, a planar inductor is inclusive of a printed circuit board (PCB) based planar inductor. A metallic shield may be used to enclose the inductor (e.g., winding structures and magnetic core) and contain its electric field to improve EMC. In this regard, the conductive or metallic shields of inductors are generally implemented using metallic enclosures external to the PCB windings and core. However, the use of such a shielding enclosure complicates manufacturing which increases cost and design complexity. Therefore, it is desirable to integrate conductive shielding within the planar power inductor structure using standard printed circuit board (PCB) manufacturing processes. In this way, the integrated shielded inductor could be adopted to improve EMC with minimal impact on system cost and complexity.

[0014] The embodiments described herein include a planar inductor implementing a winding structure using integrated shielding or conductive shields to contain the electric field of the inductor. The inductor according to the embodiments can be partially or entirely implemented using standard PCB manufacturing processes, to provide a simple and inexpensive realization. This concept effectively contains the electric field of the planar inductor, which can greatly reduce the EMI emissions when adopted in a switch-mode power converter. The planar inductor integrating conductive shields can further be designed in a way to minimize eddy currents in the shield to avoid increasing power loss. The planar inductor according to the embodiments can be implemented at a low-cost with convenient manufacturing due to its ability to be integrated with a PCB. The conductive shields can effectively contain electric fields with minimal power loss.

[0015] In the context outlined above, a planar inductor according to the embodiments includes a winding structure including a plurality of conductive windings arranged in layers of a PCB around a core. The plurality of conductive windings include at least a top winding and a bottom winding. The planar inductor also includes a conductive shield encasing the winding structure. The conductive shield includes an upper conductive shield layer disposed over the winding structure in a first direction, a lower conductive shield layer disposed under the winding structure in a second direction opposite the first direction, an inner edge shield plate disposed within the plurality of conductive windings, and an outer edge shield plate disposed around and outside the plurality of conductive windings.

[0016] Referring now to the drawings, FIG. **1** depicts a cross-sectional schematic of an example planar inductor **100** implementing conductive shielding using shield layers and conductive edge plating according to various embodiments of the present disclosure. The planar inductor **100** is illustrated as a representative example and is not drawn to any particular size or scale. The planar inductor **100** is depicted to introduce the concepts of planar (e.g., PCB-implemented) inductor structures with integrated shields. The concepts described herein can be extended to a range of different inductor structures, several of which are described below.

[0017] The planar inductor **100** shown in FIG. **1** can be implemented in a PCB with other power conversion components or can alternatively be implemented in a separate PCB and electrically coupled to other power conversion components of a power converter. For example, the planar inductor **100** can be a power factor correction (PFC) inductor used to improve the efficiency and stability of power supplies or other power converters, especially in switch-mode power supplies

(SMPS). The planar inductor **100** can be used with a variety of power converter topologies such as buck, boost, and buck-boost, among others. The planar inductor **100** can be implemented in the same PCB with semiconductor switches, gate drivers, signal processing sensors, and alternating current (AC) line EMI filters, among others. As mentioned above, the planar inductor **100** does not have to be implemented in the same PCB with these other components but it may be beneficial to implement the planar inductor **100** in the same PCB as the other components from efficiency, manufacturing, and cost perspectives.

[0018] The planar inductor **100** can be configured to contain electric fields emanating or emitted from a winding structure of the planar inductor **100** by way of a conductive shield that encases the winding structure. The conductive shield is a metallic shield that can be configured to contain the electric fields and prevent parasitic electromagnetic couplings between the planar inductor **100** and other nearby power conversion components, as mentioned above. For example, the conductive shield can prevent parasitic capacitive coupling from the planar inductor **100** to an EMI filter, thereby reducing EMI and switching noise of a power converter. Or, the conductive shield can prevent parasitic capacitive coupling from the planar inductor **100** to auxiliary gate-drive and sensing circuitry, thereby decreasing interference with these circuits and improving the reliability of a power converter. It should be noted that the planar inductor **100** and its implementations shown in FIG. 1 is not exhaustively illustrated, meaning that other components not shown in FIG. 1 can be included or relied upon in some cases. Similarly, one or more components shown in FIG. 1 can be omitted in some cases.

[0019] The planar inductor **100** includes a core **103** and an inductor structure **150**. The inductor structure **150** can be implemented using a PCB, as described below, and the core **103** can be implemented with one or multiple pieces such that it can be positioned around and, in some cases, through openings in the PCB. The inductor structure **150** includes a winding structure **160** and a conductive shield **152**. The winding structure **160** includes a plurality of conductive windings (“windings”) that can be arranged in layers of a PCB around the core **103**. The inductor structure **150** adopts or tracks a substantially or circular shape and is positioned around a central leg **105** of the core **103**. In other words, inductor structure portion **150A** and inductor structure portion **150B**, which are structurally identical, are positioned around the central leg **105**. Based on the structure of the core **103**, which is an EQ shaped core in this case, the inductor structure **150** can track different shapes such as oval, square-like, octagonal, rectangular, and hexagonal, among others. The core **103** is a magnetic core and can be embodied as a material of high magnetic permeability, such as a ferromagnetic material like iron, laminated silicon steel, laminated iron sheets, or other solid or laminated ferromagnetic ceramic, metal, metal alloy, or related material(s).

[0020] The windings of the winding structure **160** include metallic windings such as copper windings and can include an arrangement of individual windings with each winding being implemented in a separate metallic layer or trace of the PCB. In some cases, multiple windings can be implemented in the same metallic layer. In FIG. 1, the windings of the winding structure **160** include four windings implemented in four separate metallic layers of the PCB and arranged around the core **103**, but the winding structure **160** is not limited to four windings and can include less than or greater than four windings. The core **103** can include an EQ shaped core as stated above. However, the planar inductor **100** is not limited thereto, and the windings can be designed to be arranged around different types of cores such as a UI shaped core and other types of cores.

[0021] The planar inductor **100** includes the conductive shield **152**. The planar inductor **100** encloses or encases the winding structure **160**. So that the conductive shield **152** can effectively encase or completely enclose the winding structure **160**, a geometric shape of the conductive shield **152** tracks or adopts a geometric shape of the winding structure **160**. For example, if the winding structure **160** tracks a circular shape, the conductive shield **152** can track a circular shape.

[0022] The conductive shield **152** is a metallic shield and can be formed from the same materials and components as the individual windings of the winding structure **160**, such as copper or copper

traces in a PCB board. The conductive shield **152** includes an upper conductive shield layer **153**, a lower conductive shield layer **154**, an inner edge shield plate **156**, and an outer edge shield plate **158**. The upper conductive shield layer **153**, the lower conductive shield layer **154**, the inner edge shield plate **156**, and the outer edge shield plate **158** can all be electrically coupled to each other, and function to contain electric fields emanating from the planar inductor **100** to mitigate undesirable electromagnetic couplings to other components and conductors within the power converter. For example, the conductive shield **152** can function to contain electric fields emanating from the planar inductor **100** during a power conversion operation to prevent parasitic capacitive coupling from the planar inductor **100** to an EMI filter and other auxiliary circuits that can be present within a power converter such as gate drivers and sensing circuits, thereby decreasing interference with these circuitries and improving reliability of the power converter. The conductive shield **152** can be manufactured with the planar inductor **100** as an integrated metallic shield using standard PCB manufacturing processes, to improve EMC with minimal impact on system cost and complexity.

[0023] The upper conductive shield layer **153** is disposed over the winding structure **160** in a first direction. For example, the upper conductive shield layer **153** can be implemented in a metallic layer of a PCB above a metallic layer of an uppermost or a top winding of the winding structure **160**. The lower conductive shield layer **154** is disposed under the winding structure **160** in a second direction opposite the first direction. For example, the lower conductive shield layer **154** can be implemented in a metallic layer of a PCB below a metallic layer of a lowermost or a bottom winding of the winding structure **160**. The inner edge shield plate **156** is disposed within the conductive windings of the winding structure **160** and positioned adjacent to the central leg **105**. The outer edge shield plate **158** is disposed around and outside of the conductive windings of the winding structure **160** and positioned away from the central leg **105**.

[0024] The upper conductive shield layer **153** and the lower conductive shield layer **154** are substantially equal in thickness to each other and to a thickness of an individual winding of the windings of the winding structure **160**, in at least one example. The inner edge shield plate **156** and the outer edge shield plate **158** are substantially equal in thickness to each other and have thicknesses lower in value than thicknesses of the upper conductive shield layer **153** and the lower conductive shield layer **154**. The upper conductive shield layer **153** and the lower conductive shield layer **154** can be implemented or integrated with conductive metal traces, disposed vertically with respect to the top and the bottom conductive windings of the winding structure **160**, to prevent electric fields of the planar inductor **100** from escaping vertically through the PCB. The edge shield plates **156** and **158** are implemented lateral to the winding structure **160** to prevent electric fields of the planar inductor **100** escaping laterally through the PCB and can be implemented by PCB edge plating process or other means. Further description regarding the conductive shield **152** is provided below with respect to FIGS. 2 and 3.

[0025] FIG. 2 depicts a cross-sectional PCB schematic of an example inductor structure implementing the conductive shield shown in FIG. 1, and FIG. 3 depicts a perspective view of a planar inductor implementing the inductor structure shown in FIG. 2, according to various embodiments of the present disclosure. An inductor structure **250** is similar to the inductor structure portions **150A** or **150B** but includes six conductive windings **220**, **222**, **224**, **226**, **228**, and **230** (“windings **220-230**”), as parts of a winding structure **260**, instead of four windings in the inductor structure **150**. The inductor structure **250** can be implemented in the planar inductor **100** shown in FIG. 1. The inductor structure **250** includes the conductive shield **152**, which encases or completely encloses the winding structure **260**.

[0026] The upper conductive shield layer **153** is disposed over the winding structure **260** in a first direction. For example, the upper conductive shield layer **153** can be implemented in a metallic layer of a PCB above the winding **220**, which is a top winding of the winding structure **260**. The lower conductive shield layer **154** is disposed under the winding structure **260** in a second direction

opposite the first direction. For example, the lower conductive shield layer **154** can be implemented in a metallic layer of a PCB below the winding **230**, which is a bottom winding of the winding structure **260**. The inner edge shield plate **156** is disposed within and around the windings **220-230** of the winding structure **260** and can be positioned adjacent to the central leg **105**. The outer edge shield plate **158** is disposed around and outside of the windings **220-230** of the winding structure **260** and can be positioned away from the central leg **105**.

[0027] The windings **220-230** are separated from the inner edge shield plate **156** by an edge clearance **272**. The windings **220-230** are separated from each other by a dielectric thickness **276**. Each of the windings **220-230** has a winding thickness **270**, which can vary based on manufacturer specifications or based on the application of the inductor structure **250**. In one example, the winding thickness **270** can range from 17 microns to 140 microns, commensurate with about a half ounce (oz) copper to four oz copper for each of the windings **220-230**. The inner edge shield plate **156** and the outer edge shield plate **158** each has an edge shield plate thickness **278**, which is less than the winding thickness **270**. The conductive shield **152** has a winding window **213**, which defines the total cross-sectional width of the inductor winding structure **250** including windings, conductive shielding, and associated clearances.

[0028] Referring to FIG. 3, a planar inductor **300** tracks a circular shape and can include the inductor structure **250** or the inductor structure **150**. The planar inductor **300** is similar to the planar inductor **100** and can be positioned around the central leg **105** of the core **103**. The inner edge shield plate **156** is disposed lateral to the winding structure **260**, which is implemented in the planar inductor **300**. The inner edge shield plate **156** extends circularly within the winding structure **260** and forms an inner circumference **382** of the conductive shield **152**. The outer edge shield plate **158** is disposed lateral to the winding structure **260**, extends circularly around and outside the winding structure **260**, and forms an outer circumference **380** of the conductive shield **152**. The conductive shield **152** encases or completely encloses the winding structure **260** to contain the electric fields emanating from the planar inductor **300**.

[0029] The conductive shield **152** includes a termination **310** which can be used to connect to an input voltage ($V_{sub.in}$) or output voltage ($V_{sub.o}$) terminal such as a direct current (DC) or alternating current (AC) terminal of a power converter. The conductive shield **152** includes a separation or a cut **390** that extends completely from an edge of the outer circumference **380** to an edge of the inner circumference **382** including separation of the inner edge shield plate **156**, the outer edge shield plate **158**, the upper conductive shield layer **153**, and the lower conductive shield layer **154**. The cut **390** physically separates the conductive shield **152** into two parts with respect to the termination **310** to prevent eddy currents and circulation of large current that could potentially flow around the loop formed by the conductive shield **152**, which could otherwise degrade or destroy its performance.

[0030] The upper conductive shield layer **153** and the lower conductive shield layer **154** are generally equal in thickness to each other and may be equal, greater than, or less than the thickness of an individual winding of the windings of the winding structure **160**. The inner edge shield plate **156** and the outer edge shield plate **158** are generally equal in thickness to each other and have thicknesses lower in value than thicknesses of the upper conductive shield layer **153** and the lower conductive shield layer **154**. The upper conductive shield layer **153** and the lower conductive shield layer **154** can be implemented or integrated with conductive metal traces vertically with respect to the top and the bottom conductive windings of the winding structure **160**, to prevent electric fields of the planar inductor **100** from escaping vertically from the PCB. The edge shield plates **156** and **158** can be implemented or integrated with conductive edge plating implemented lateral to the winding structure **160** to prevent electric fields of the planar inductor **100** escaping laterally through the PCB.

[0031] FIG. 4 depicts a cross-sectional schematic of an example planar inductor with an alternative implementation of a conductive shield using conductive shield rings located co-planar with

inductor windings, according to various embodiments of the present disclosure. A planar inductor **400** can be implemented in a PCB with other power conversion components or can alternatively be implemented in a separate PCB and electrically coupled to other power conversion components of a power converter, such as a buck, boost, or buck-boost converter. The planar inductor **400** is not exhaustively illustrated, meaning that other components not shown in FIG. 4 can be included or relied upon in some cases. Similarly, one or more components shown in FIG. 4 can be omitted in some cases. The planar inductor **400** is similar or substantially identical to the planar inductor **100** but includes multiple conductive inner shield rings and outer shield rings located co-planar with inductor windings of a winding structure, in contrast with the solid conductive edge plating used for the inner edge plate **156** and the outer edge plate **158** shown in FIG. 1.

[0032] The planar inductor **400** includes a core **403** and an inductor structure **450**, which includes a winding structure **460** and a conductive shield **452**. The winding structure **460** includes a plurality of conductive windings (“windings”) that can be arranged in layers of a PCB around the core **403**. The inductor structure **450** adopts or tracks a substantially circular shape and is positioned around a central leg **405** of the core **403**. In other words, inductor structure portion **450A** and inductor structure portion **450B**, which are structurally identical, are positioned around the central leg **405**. Based on the structure of the core **403**, which is an EQ shaped core in this case, the inductor structure **450** can track different shapes such as oval, square-like, octagonal, and hexagonal, among others.

[0033] The windings of the winding structure **460** include metallic windings such as copper windings and can include an arrangement of individual windings with each winding implemented in a separate metallic layer or trace of the PCB. In some cases, multiple windings can be implemented in the same metallic layer. In FIG. 4, the windings of the winding structure **460** include four windings implemented in four separate metallic layers of the PCB and arranged around the core **403**, but the winding structure **460** is not limited to four windings and can include less than or greater than four windings. The core **403** can include an EI shaped core or an EQ shaped core as stated above. However, the planar inductor **400** is not limited thereto, and the windings can be designed to be arranged around different types of cores such as a UI shaped core and other types of cores.

[0034] The conductive shield **452** encloses or encases the winding structure **460**. So that the conductive shield **452** can effectively encase or completely enclose the winding structure **160**, a geometric shape of the conductive shield **452** tracks or adopts a geometric shape of the winding structure **460**. For example, if the winding structure **160** tracks a circular shape, the conductive shield **452** can track a circular shape. The conductive shield **452** includes an upper conductive shield layer **453**, a lower conductive shield layer **454**, inner conductive shield rings **456**, and outer conductive shield rings **458**. The upper conductive shield layer **453**, the lower conductive shield layer **454**, the inner conductive shield rings **456**, and the outer conductive shield rings **458** can all be electrically coupled to each other, and function to contain electric fields emanating from the planar inductor **100** to mitigate undesirable electromagnetic couplings to other components and conductors within a power converter. For example, the conductive shield **452** can function to contain electric fields emanating from the planar inductor **400** during a power conversion operation to prevent parasitic capacitive coupling from the planar inductor **400** to an EMI filter and other auxiliary circuits that can be present within a power converter such as gate drivers and sensing circuits, thereby decreasing interference with these circuitries and improving reliability of the power converter. The conductive shield **452** can be manufactured with the planar inductor **400** as an integrated metallic shield using standard PCB manufacturing processes, to improve EMC with minimal impact on system cost and complexity.

[0035] The upper conductive shield layer **453** is disposed over the winding structure **460** in a first direction. For example, the upper conductive shield layer **453** can be implemented in a metallic layer of a PCB above a metallic layer of an uppermost or a top winding of the winding structure

460. The lower conductive shield layer **454** is disposed under the winding structure **460** in a second direction opposite the first direction. For example, the lower conductive shield layer **454** can be implemented in a metallic layer of a PCB below a metallic layer of a lowermost or a bottom winding of the winding structure **460**. The inner conductive shield rings **456** are implemented co-planar and on the same metallic layers as the winding structure **460** and adjacent to central leg **405**. The outer conductive shield rings **458** are implemented co-planar and on the same metallic layers as the winding structure **460** and around and outside of the conductive windings of the winding structure **460** and positioned away from the central leg **405**.

[0036] In contrast to the upper conductive shield layer **153** and the lower conductive shield layer **154** of the conductive shield **152**, the upper conductive shield layer **453** and the lower conductive shield layer **454** are patterned with radial cuts. In contrast to the inner edge shield plate **156** and/or the outer edge shield plate **158**, which includes a single conductive plate, the inner conductive shield rings **456** and the outer conductive shield rings **458** each include separated conductive traces disposed around the winding structure **460**. The structure of the conductive shield **452** is discussed in greater detail with respect to FIGS. **6** and **7**, provided below.

[0037] FIG. **5** depicts a cross-sectional PCB schematic of an example inductor structure implementing the conductive shield shown in FIG. **4**, FIG. **6** depicts a perspective view of a planar inductor implementing the inductor structure shown in FIG. **5**, and FIG. **7** depicts a sectional view of the planar inductor shown in FIG. **6**, according to various embodiments of the present disclosure. Referring to FIG. **5**, an inductor structure **550** is similar to the inductor structure portions **450A** or **450B** but includes a winding structure **560** with six conductive windings instead of the four windings in the inductor structure **450**. The inductor structure **550** can be implemented in the planar inductor **400** shown in FIG. **4**. The inductor structure **550** includes the conductive shield **452**, which encases or completely encloses the winding structure **560**.

[0038] The upper conductive shield layer **453** is disposed over the winding structure **560** in a first direction. For example, the upper conductive shield layer **453** can be implemented in a metallic layer of a PCB above an uppermost or top winding of the winding structure **560**. The lower conductive shield layer **454** is disposed under the winding structure **560** in a second direction opposite the first direction. For example, the lower conductive shield layer **454** can be implemented in a metallic layer of a PCB below a lowermost or bottom winding of the winding structure **560**. The inner conductive shield rings **456** are disposed within and around the winding structure **560** and can be positioned adjacent to the central leg **405**. The outer conductive shield rings **458** are disposed around and outside of the winding structure **560** and can be positioned away from the central leg **405**.

[0039] The winding structure **560** is separated from the inner conductive shield rings **456** and the outer conductive shield rings **458** by an edge clearance **572**. The windings in the winding structure **560** are separated from each other by a dielectric thickness **576**. Additionally, the conductive traces of the inner conductive shield rings **456** and the outer conductive shield rings **458** are separated from each other by the same dielectric thickness **576**. Each of the windings in the winding structure **560**, the upper conductive shield layer **453**, the lower conductive shield layer **454**, and each conductive trace of the inner conductive shield rings **456** and the outer conductive shield rings **458** has a winding thickness **570** in at least one example which can vary based on manufacturer specifications or based on the application of the inductor structure **550**. The thickness of the upper conductive shield layer **453** is generally equal to the thickness of lower conductive shield layer **454**. The thickness of individual windings of the winding structure **560**, the inner conductive shield rings **456**, and the outer conductive shield rings **458** is generally equal to, but may be less than or greater than, the thickness of the upper conductive shield layer **453** and lower conductive shield layer **454**. In one example, the winding thickness **570** of each of the windings in the winding structure **560**, the upper conductive shield layer **453**, the lower conductive shield layer **454**, and each conductive trace of the inner conductive shield rings **456** and the outer conductive shield rings

458 can range from 17 microns to 140 microns, commensurate with about a half ounce (oz) copper to four oz copper. Each conductive trace of the inner conductive shield rings **456** and the outer conductive shield rings **458** has a width **574**. The inner conductive shield rings **456** and the outer conductive shield rings **458** form a shield around inner and outer circumference formed by the winding structure **560**. The conductive shield **452** has a winding window **513**, which defines the total cross-sectional width of the inductor winding structure **550** including windings, conductive shielding, and associated clearances.

[0040] The upper conductive shield layer **453**, the lower conductive shield layer **454**, and each conductive trace of the inner conductive shield rings **456** and the outer conductive shield rings **458** may or may not be equal in thicknesses. The winding structure **560** includes six conductive windings in the example shown, although the winding structure **560** can include greater or fewer than six windings. The outer conductive shield rings **458** include the same number of separated conductive traces as the total number of windings of the winding structure **560**. Similarly, the inner conductive shield rings **456** includes the same number of separated conductive traces as the total number of windings of the winding structure **560**. Thus, in the example shown, the outer conductive shield rings **458** and the inner conductive shield rings **456** both include six separated conductive traces.

[0041] The outer conductive shield rings **458** are aligned with the winding structure **560**. That is, the separated conductive traces of the outer conductive shield rings **458** are aligned with the windings of the winding structure **560**. Similarly, the inner conductive shield rings **456** are aligned with the winding structure **560**. That is, the separated conductive traces of the inner conductive shield rings **456** are aligned with the windings of the winding structure **560**. The upper conductive shield layer **453** and the lower conductive shield layer **454** help prevent electric fields of the planar inductor **400** from escaping vertically from the PCB. The inner conductive shield rings **456** and the outer conductive shield rings **458** help prevent electric fields of the planar inductor **400** escaping laterally through the PCB.

[0042] In FIG. 6, a planar inductor **600** is implemented in a PCB **615**. The planar inductor **600** includes the conductive shield **452** and is representative of the inductor structure **450** or **550**. The following description is provided with respect to the inductor structure **550**, but this description is applicable to the inductor structure **450** as well. As the winding structure **560** tracks a circular shape, the conductive shield **452** also tracks a circular shape to enclose the winding structure **560**. The inner conductive shield rings **456** are disposed within and around the winding structure **560** and form an inner circumference **682** of the conductive shield **452**. The outer conductive shield rings **458** are disposed around and outside the winding structure **560** and form an outer circumference **680** of the conductive shield **452**. An intermediary section **684** of the upper conductive shield layer **453** forms a loop between the outer circumference **680** and the inner circumference **682**. The intermediary section **684** is approximately halfway between the outer circumference **680** and the inner circumference **682**. The lower conductive shield layer **454** includes a similar intermediary section that is approximately halfway between the outer circumference **680** and the inner circumference **682**.

[0043] Similar to the conductive shield **152**, the conductive shield **452** includes a termination (not depicted) which can be used to connect to a DC or AC V.sub.o terminal of a power converter. The conductive shield **452** similarly includes a separation or cut (not depicted) that extends from an edge of the outer circumference **680** to an edge of the inner circumference **682**. The cut physically separates the conductive shield **452** into two sections with respect to a shield termination to prevent eddy currents and circulation of large current that could potentially flow around the loop formed by the conductive shield **452**, which could otherwise degrade or destroy its performance.

[0044] Referring to FIG. 7, the upper conductive shield layer **453** is patterned with radial cuts extending from the outer circumference **680** or the inner circumference **682** to the intermediary section **684**. The lower conductive shield layer **454** is patterned with similar radial cuts extending

from the outer circumference **680** or the inner circumference **682** to the intermediary section of the lower conductive shield layer **454**. The radial cuts help minimize eddy currents in the conductive shield **452** and form a fishbone-like pattern for the conductive shield **452**. It should be noted that the conductive shield **452** can include other types of patterns and not just radial cuts as described above. For example, other cut patterns can be applied to the upper conductive shield layer **453** and the lower conductive shield layer **454** as long as cuts or separations are present in each of the shield layers **453** and **454**.

[0045] It should be noted that the planar inductors according to the embodiments can include inductor structures that are combinations of the inductor structures **150** and **450**. For example, a planar inductor according to the embodiments can include solid upper and lower conductive shield layers (e.g., the upper conductive shield layer **153** and the lower conductive shield layer **154**) that are not patterned and also use conductive inner and outer shield rings with separated conductive traces (e.g., the inner conductive shield rings **456** and the outer conductive shield rings **458**). In another example, a planar inductor according to the embodiments can include patterned upper and lower conductive shield layers (e.g., the upper conductive shield layer **453** and the lower conductive shield layer **454**) and solid inner and outer edge shield plates (e.g., the inner edge shield plate **156** and the outer edge shield plate **158**).

[0046] FIG. **8** depicts a planar inductor **800** with a patterned conductive shield implemented with a UI shaped core according to various embodiments of the present disclosure. The planar inductor **800** includes the inductor structure **550** which is implemented with a UI shaped core **603**. The core **603** is a magnetic core with a plurality of core legs. In particular, the inductor structure **550** is positioned around one core leg of the plurality of core legs of the core **603**. Although a UI shaped core is depicted, the planar inductor **800** can include different types of cores such as EQ and EI shaped cores, among others types of cores, in which the inductor structure **550** can be implemented with. The core **603** is provided as a representative example and can be extended to include any number of legs, as needed, based on the number of inductor structures needed for the planar inductor **800**. The core **603** can be embodied as a material of high magnetic permeability, such as a ferromagnetic material like iron, laminated silicon steel, laminated iron sheets, or other solid or laminated ferromagnetic ceramic, metal, metal alloy, or related material(s).

[0047] The described embodiments include a planar inductor structure incorporating conductive shields integrated within a PCB to contain the inductor's electric field to mitigate undesirable electromagnetic couplings to other components and conductors within a power converter. This shielded structure can improve the EMC and reliability of the power converter by decreasing interference with these other circuitries and components. In one embodiment, the conductive shield can be implemented using solid shield layers and PCB edge plating to form a complete shield around the conductive windings of the winding structure. To reduce loss generated by eddy currents in the shield, the conductive shields of the embodiments can also be implemented using patterned shield layers and conductive inner and outer shield rings that form a shield around the inner and outer circumferences of the winding structure. The planar inductors of the embodiments effectively contain the electric field produced by the inductor to prevent parasitic capacitive couplings to other areas of the converter circuit.

[0048] The planar inductors (e.g., the planar inductors **100**, **300**, **400**, and **600**), when implemented in practice, will generally include PCB dielectric material as part of the PCB inductor structure. This PCB dielectric material is not specifically illustrated or described in detail herein for the sake of clarity for the inductor structure. Additionally, the planar inductors can include magnetic components having more than one set of conductive windings, such as coupled inductors and transformers. For example, the conductive shields (e.g., the conductive shields **152** or **452**) of the embodiments and their concepts can be applied to winding components of transformers. In other words, the conductive shields can be extended to encase or enclose the winding components of transformers and provide similar system benefits in terms of EMC and reliability as what was

described with respect to the inductors herein. Further, the conductive shields of the embodiments can be applied to a variety of other shapes of windings and winding structures, such as rectangular, oval, square-like, and hexagonal, among others. The planar inductors of the embodiments can also include or be implemented with other shapes of magnetic cores, such as E-cores, U-cores, I-cores, EE-cores, UI cores, EQ cores, EI cores, UI cores, and EE cores, among other types of cores. The structures and shapes provided in the embodiments are provided as examples only and can be extended to encompass a wide range of other shapes and structures, such as the ones described above, while staying within the concepts and principles described in the disclosure.

[0049] When adopted in a switch-mode power converter, one implementation of the planar inductor of the embodiments is demonstrated to reduce a converter's common mode noise emissions by up to 25 dBV. Meanwhile, the efficiency of the converter adopting one or more of the planar inductors of the embodiments has an efficiency only 0.14% lower than an equivalent converter using a planar inductor with no shielding. The embodiments can be implemented for a wide range of planar inductor designs with various winding patterns and magnetic core geometries. The embodiments provide an attractive means to improve the EMC and reliability of switching power converters in a wide range of applications, such as offline power supplies, electric vehicle chargers, and grid-tied converters.

[0050] The features, structures, or characteristics described above may be combined in one or more embodiments in any suitable manner, and the features discussed in the various embodiments are interchangeable, if possible. In the following description, numerous specific details are provided in order to fully understand the embodiments of the present disclosure. However, a person skilled in the art will appreciate that the technical solution of the present disclosure may be practiced without one or more of the specific details, or other methods, components, materials, and the like may be employed. In other instances, well-known structures, materials, or operations are not shown or described in detail to avoid obscuring aspects of the present disclosure.

[0051] Although the relative terms such as “on,” “below,” “upper,” and “lower” are used in the specification to describe the relative relationship of one component to another component, these terms are used in this specification for convenience only, for example, as a direction in an example shown in the drawings. It should be understood that if the device is turned upside down, the “upper” component described above will become a “lower” component. When a structure is “on” another structure, it is possible that the structure is integrally formed on another structure, or that the structure is “directly” disposed on another structure, or that the structure is “indirectly” disposed on the other structure through other structures.

[0052] In this specification, the terms such as “a,” “an,” “the,” and “said” are used to indicate the presence of one or more elements and components. The terms “comprise,” “include,” “have,” “contain,” and their variants are used to be open ended, and are meant to include additional elements, components, etc., in addition to the listed elements, components, etc. unless otherwise specified in the appended claims. If a component is described as having “one or more” of the component, it is understood that the component can be referred to as “at least one” component.

[0053] The terms “first,” “second,” etc. are used only as labels, rather than a limitation for a number of the objects. It is understood that if multiple components are shown, the components may be referred to as a “first” component, a “second” component, and so forth, to the extent applicable.

[0054] Disjunctive language such as the phrase “at least one of X, Y, or Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to present that an item, term, etc., can be either X, Y, or Z, or any combination thereof (e.g., X; Y; Z; X or Y; X or Z; Y or Z; X, Y, or Z; etc.). Thus, such disjunctive language is not generally intended to, and should not, imply that certain embodiments require at least one of X, at least one of Y, or at least one of Z to each be present.

[0055] The terms “about” and “substantially,” unless otherwise defined herein to be associated with a particular range, percentage, or related metric of deviation, account for at least some

manufacturing tolerances between a theoretical design and manufactured product or assembly, such as the geometric dimensioning and tolerancing criteria described in the American Society of Mechanical Engineers (ASME®) Y14.5 and the related International Organization for Standardization (ISO®) standards. Such manufacturing tolerances are still contemplated, as one of ordinary skill in the art would appreciate, although “about,” “substantially,” or related terms are not expressly referenced, even in connection with the use of theoretical terms, such as the geometric “perpendicular,” “orthogonal,” “vertex,” “collinear,” “coplanar,” and other terms.

[0056] The above-described embodiments of the present disclosure are merely possible examples of implementations set forth for a clear understanding of the principles of the disclosure. Many variations and modifications may be made to the above-described embodiment(s) without departing substantially from the spirit and principles of the disclosure. All such modifications and variations are intended to be included herein within the scope of this disclosure and protected by the following claims.

Claims

1. A planar inductor, comprising: a winding structure comprising a plurality of conductive windings arranged in layers of a printed circuit board (PCB) around a core, the plurality of conductive windings comprising at least a top winding and a bottom winding; and a conductive shield encasing the winding structure, the conductive shield comprising: an upper conductive shield layer disposed over the winding structure in a first direction; a lower conductive shield layer disposed under the winding structure in a second direction opposite the first direction; an inner conductive shield disposed within and around the plurality of conductive windings; and an outer conductive shield disposed around and outside the plurality of conductive windings.
2. The planar inductor of claim 1, wherein a geometric shape of the conductive shield tracks a geometric shape of the winding structure.
3. The planar inductor of claim 1, wherein a thickness of the upper conductive shield layer is different than a thickness of the inner conductive shield or the outer conductive shield.
4. The planar inductor of claim 1, wherein: the upper conductive shield layer, the lower conductive shield layer, the inner conductive shield, and the outer conductive shield are electrically coupled to each other; and the conductive shield is electrically coupled to an input or output voltage potential of a power converter.
5. The planar inductor of claim 1, wherein the conductive shield is configured to mitigate or contain an electric field emanating from the winding structure during a power conversion operation.
6. The planar inductor of claim 1, wherein one of the upper or the lower conductive shield layers is patterned with a plurality of radial cuts.
7. The planar inductor of claim 6, wherein the plurality of radial cuts extends from an outer circumference of the conductive shield to an intermediary section of the upper or the lower conductive shield layers.
8. The planar inductor of claim 6, wherein: the inner conductive shield comprises a plurality of separate conductive traces disposed around and within the plurality of conductive windings; and the outer conductive shield comprises a plurality of separate conductive traces disposed around and outside the plurality of conductive windings.
9. The planar inductor of claim 1, wherein the inner conductive shield and the outer conductive shield are uniform structures without variation in surface characteristics.
10. The planar inductor of claim 1, wherein the upper conductive shield layer and the lower conductive shield layer are uniform structures without variations in surface characteristics.
11. The planar inductor of claim 1, wherein a thickness of the upper conductive shield layer is equal to, greater than, or less than a thickness of the inner conductive shield or the outer conductive

shield.

12. A planar inductor, comprising: a winding structure comprising a plurality of conductive windings arranged in layers of a printed circuit board (PCB) around a core, the plurality of conductive windings comprising at least a top winding and a bottom winding; and a patterned conductive shield encasing the winding structure, the patterned conductive shield comprising: an upper conductive shield layer disposed over the winding structure in a first direction, the upper conductive shield layer comprising a plurality of radial cuts; a lower conductive shield layer disposed under the winding structure in a second direction opposite the first direction, the lower conductive shield layer comprising a plurality of radial cuts; an inner conductive shield disposed within and around the plurality of conductive windings; and an outer conductive shield disposed around and outside the plurality of conductive windings.

13. The planar inductor of claim 12, wherein the plurality of radial cuts of the upper conductive shield layer extends from an outer circumference of the patterned conductive shield to an intermediary section of the upper conductive shield layer.

14. The planar inductor of claim 12, wherein the plurality of radial cuts of the lower conductive shield layer extends from an inner circumference of the patterned conductive shield to an intermediary section of the lower conductive shield layer.

15. The planar inductor of claim 13, wherein: the outer conductive shield comprises a plurality of separate conductive traces disposed around and outside the plurality of conductive windings; and the outer conductive shield forms the outer circumference of the patterned conductive shield.

16. The planar inductor of claim 14, wherein: the inner conductive shield comprises a plurality of separate conductive traces disposed within and around the plurality of conductive windings; and the inner conductive shield forms the inner circumference of the patterned conductive shield.

17. The planar inductor of claim 12, wherein a thickness of the upper conductive shield layer is equal to a thickness of the lower conductive shield layer.

18. The planar inductor of claim 12, wherein: the upper conductive shield layer, the lower conductive shield layer, the inner conductive shield, and the outer conductive shield are electrically coupled to each other; and the patterned conductive shield is electrically coupled to an output voltage potential of a power converter.

19. The planar inductor of claim 12, wherein a thickness of the lower conductive shield layer is equal to, greater than, or less than a thickness of the inner conductive shield or a thickness of the outer conductive shield.

20. The planar inductor of claim 12, wherein the core is an EQ shaped core or a UI shaped core.
